

Generalized Joint Laxity Is an Independent Predictor of Postoperative Resprain Following Anatomic Reconstruction Using Gracilis Tendon Autograft for Patients With Chronic Lateral Ankle Instability

Haoxuan Liu,* MD, Jianhong Li,[†] MD, Tong Su,* MD, Yanfang Jiang,* MSc, Yanbin Pi,* MD, Xing Xie,* MD, Qinwei Guo,* MD , Chen Jiao,*[‡] MD, and Dong Jiang,*[‡] MD 
Investigation performed at Peking University Third Hospital, Beijing, China

Background: Generalized joint laxity (GJL) has been reported as a risk factor for poor outcomes following multiligament surgery. However, no previous studies have investigated whether GJL is a risk factor of postoperative resprain following anatomic reconstruction using gracilis tendon autograft.

Purpose: To investigate whether GJL is a risk factor of postoperative resprain in chronic lateral ankle instability (CLAI) patients following anatomic reconstruction using gracilis tendon autograft.

Study Design: Cohort study; Level of evidence, 3.

Methods: A total of 290 consecutive CLAI patients who underwent anatomic reconstruction using gracilis tendon autograft from January 2018 to September 2023 with a minimum follow-up of 24 months were retrospectively evaluated and were divided into 2 groups according to the presence of GJL (based on age and Beighton score). Clinical outcomes were measured through the presence of postoperative noncontact resprain and recurrent instability, return-to-sport status, and patient-reported outcomes including the visual analog scale (VAS) for pain, Tegner Activity Level scale, and Karlsson-Peterson ankle score (Karlsson score). Risk factors associated with postoperative noncontact resprain were determined using univariate and multivariate logistic regression analysis. Survival outcomes were compared using Kaplan-Meier analysis.

Results: A total of 72 patients were included in the GJL group (mean age at surgery, 27.5 ± 9.0 years; male, $n = 27$; female, $n = 45$), and 218 patients were included in the non-GJL group (mean age at surgery, 32.8 ± 10.2 years; male, $n = 129$; female, $n = 89$). The overall noncontact resprain rate was 6.6%, and the overall recurrent instability rate was 1.7%. There was no significant difference in the postoperative VAS, Tegner, return to recreational sports rate, return to preinjury sports rate, and other surgical complications between the 2 groups. The patients with GJL demonstrated a significantly higher postoperative noncontact resprain rate (15.3% vs 3.7%; $P = .001$), a significantly higher postoperative recurrent instability rate (6.9% vs 0.0%; $P = .001$), and a significantly lower postoperative Karlsson score (90.0 ± 6.2 vs 95.6 ± 4.4 ; $P < .001$) compared with those without GJL. Multivariate logistic regression analysis revealed that GJL was an independent predictor of postoperative noncontact resprain following anatomic reconstruction using gracilis tendon autograft ($P = .004$; odds ratio = 4.293; 95% CI, 1.612–11.431). The cumulative success rates of the patients with GJL were significantly lower than those of the patients without GJL ($P = .001$).

Conclusion: GJL is an independent predictor of postoperative noncontact resprain of CLAI patients following anatomic reconstruction using gracilis tendon autograft.

Keywords: chronic lateral ankle instability; anatomic reconstruction; autograft; generalized joint laxity; resprain

Lateral ankle sprain is one of the most prevalent sports injuries.⁴² Due to the injuries of anterior talofibular ligament (ATFL) and calcaneofibular ligament (CFL), 20% to 30% of the individuals can suffer from chronic lateral ankle instability (CLAI), which is characterized as recurrent sprain, sense of giving way, persistent pain, deficits of postural control, and muscle strength.³⁷ Surgical intervention can be considered after rigorous conservative management has failed.¹ To date, the regarded gold standard surgery for CLAI patients is the modified Broström procedure.³⁹ However, for CLAI patients with poor ligament tissue quality, traditional repair technique is usually not applicable.^{23,24,44} In such cases, anatomic reconstruction with autograft or allograft emerges as the preferred surgical alternative, providing enhanced ankle stability.^{14,33,36}

Nevertheless, some patients still experience recurrent sprains after surgery, making the management of these patients particularly challenging, yet current evidence cannot definitively determine if distinct patient characteristics necessitate extra postoperative caution or tailored treatment approaches. Therefore, identifying risk factors for resprains is crucial for predicting outcomes and adopting more personalized treatment strategies. Among them, generalized joint laxity (GJL) is an important factor that requires further investigation. Because of collagen synthesis impairment, previous research suggests that patients with GJL may experience altered biological healing.^{17,19} Multiple studies indicate that GJL can lead to inferior clinical outcomes and graft laxity in patients following anterior cruciate ligament (ACL) or posterior cruciate ligament reconstruction.^{15,16,18} Similarly, after ankle ligament reconstruction using gracilis tendon autograft, patients may also experience increased postoperative laxity due to GJL, resulting in resprain. Nevertheless, we are not aware of any studies that have evaluated the impact of GJL on the clinical outcomes of CLAI patients following anatomical reconstruction using gracilis tendon autograft.

Therefore, the current study was designed to investigate the effect of GJL on the postoperative noncontact resprain rate on CLAI patients following anatomic reconstruction using gracilis tendon autograft. It was hypothesized that GJL is an independent risk factor of postoperative resprain.

METHODS

Patient Enrollment

This was a single-center retrospective cohort study approved by our institutional review board. The CLAI patients following anatomic reconstruction using gracilis tendon autograft were enrolled after meeting all of the following criteria^{10,12}: (1) age ≥ 18 years old and ≤ 60 years old; (2) a traumatic history of ≥ 1 significant ankle sprain, and the most recent ankle sprain occurred >3 months prior to study participation; (3) persistent clinical symptoms including pain, swelling, a sense of giving way, feeling of instability, or recurrent sprain for ≥ 3 months; (4) positive anterior drawer test with increased translation of >3 mm compared with the contralateral side, and chronic ATFL and CFL injury confirmed by magnetic resonance imaging (MRI); (5) failure of rigorous conservative treatment of ≥ 6 months; and (6) arthroscopic exploration using a probe that confirmed the ATFL and/or CFL had poor tension or was missing and could not be repaired that then underwent anatomic reconstruction using gracilis tendon autograft of the ATFL and/or CFL. Patients were excluded if they fulfilled the following criteria: (1) combined bilateral ankle instability or other lower limb musculoskeletal injuries; (2) combined central and peripheral neuromuscular disease; (3) progressing or early-stage ankle arthritis (Takakura arthritis classification stage ≥ 3); and (4) unwilling to participate.

From January 2018 to September 2023, a total of 340 consecutive CLAI patients were enrolled in the study. Patients who did not meet all the criteria or had a follow-up duration <2 years were excluded, and 290 patients were included in the final analysis. GJL was assessed using the Beighton 9-point scoring system, and patients were classified as having GJL when the Beighton score was ≥ 6 points in patients <18 years old, ≥ 5 points in patients 18 to 50 years old, and ≥ 4 points in patients >50 years old.²⁶ At our institution, all patients undergoing ankle ligament surgery received a preoperative physical examination by their respective attending surgeons preoperatively to assess for GJL, and the specific Beighton score was recorded in the medical records. Based on the presence of GJL, 72 patients were included in the GJL

[‡]Address correspondence to Dong Jiang, MD, Department of Sports Medicine, Peking University Third Hospital, Institute of Sports Medicine of Peking University, Beijing Key Laboratory of Sports Injuries, Engineering Research Center of Sports Trauma Treatment Technology and Devices, Ministry of Education, No. 49 North Garden Road, Beijing 100191, China (email: Dong_jiang@bjmu.edu.cn); and Chen Jiao, Department of Sports Medicine, Peking University Third Hospital, Institute of Sports Medicine of Peking University, Beijing Key Laboratory of Sports Injuries, Engineering Research Center of Sports Trauma Treatment Technology and Devices, Ministry of Education, No. 49 North Garden Road, Beijing 100191, China (email: jiaoshuj@mail.tsinghua.edu.cn).

*Department of Sports Medicine, Peking University Third Hospital, Institute of Sports Medicine of Peking University, Beijing Key Laboratory of Sports Injuries, Engineering Research Center of Sports Trauma Treatment Technology and Devices, Ministry of Education, Beijing, China.

[†]The TCM hospital of Longquanyi, Department of Joint Movement Medicine, Chengdu, Longquanyi District, Chengdu, Sichuan, China.

H.L. and J.L. share first authorship of this article.

Final revision submitted May 28, 2026; accepted May 31, 2026.

The authors declared that they have no conflicts of interest in the authorship and publication of this contribution.

Ethical approval for this study was obtained from the medical committee institutional review board (IRB) of Peking University Third Hospital (No. IRB00006761-M20250961).

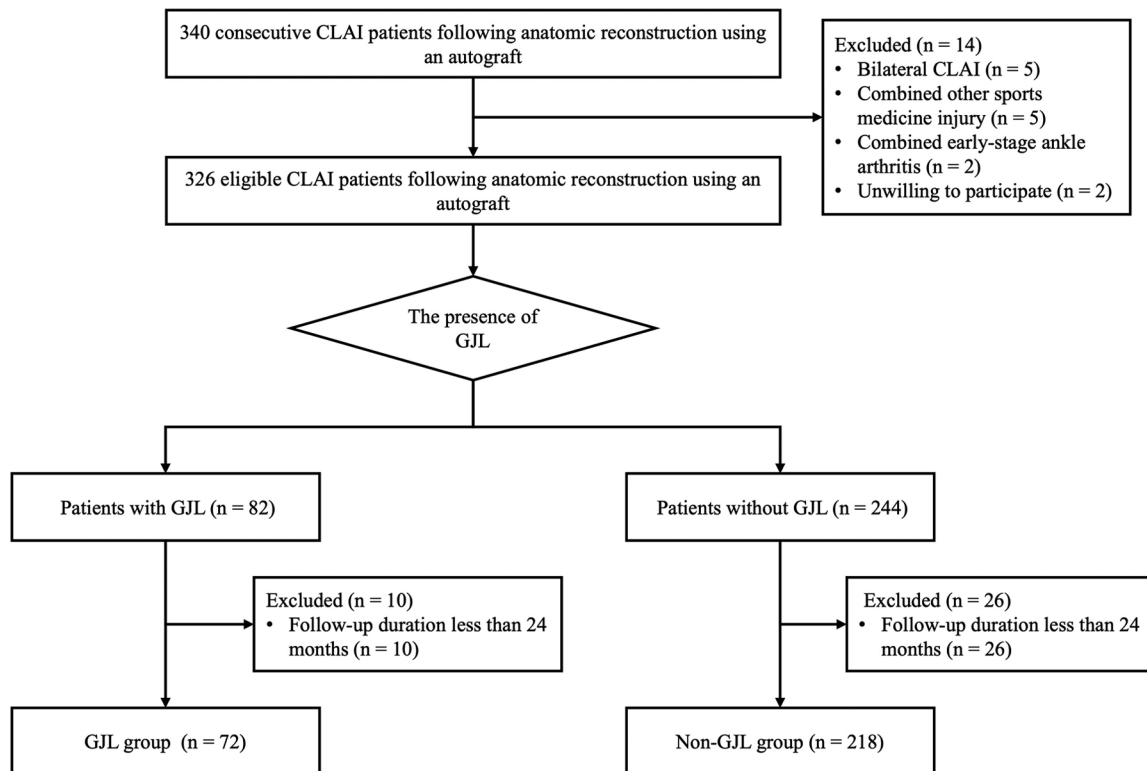


Figure 1. Patient enrollment. CLAI, chronic lateral ankle instability; GJL, generalized joint laxity.

group and 218 patients were included in the non-GJL group (Figure 1).

Surgical Procedure

All anatomic reconstructions utilizing gracilis tendon autograft for patients with CLAI were conducted by 4 surgeons (D.J., C.J., Y.P., X.X.) in accordance with established methods described in prior studies.^{35,36} Procedures were performed under either spinal or general anesthesia. Patients were positioned supine with the ipsilateral hip elevated by approximately 30°. Standard anteromedial and anterolateral arthroscopic portals were established. Diagnostic arthroscopy was performed using a 2.9-mm 30° arthroscopic system (Smith & Nephew) to evaluate and address any intra-articular pathologies, such as osteochondral lesions, osteophytes, or loose bodies. The Outerbridge classification was applied to evaluate the osteochondral lesions. For those with grade 1 to grade 3 lesions, the cartilage lesions were debrided with a mechanical shaver (Smith & Nephew). For patients with grade 4 lesions, debridement was followed by microfracture at the intact subchondral bone. The syndesmosis width was also recorded using the method reported in previous studies.³⁴ Then, the ipsilateral gracilis tendon was harvested and prepared by doubled folding. A 4- to 5-cm oblique incision was made anteroinferior to the fibula. When the remnant of the CFL was of good quality or remained intact, it was either repaired or left in situ.

Following debridement of the ATFL footprints, 2 bone tunnels were drilled for an isolated ATFL reconstruction using the prepared tendon graft. Then, the CFL remnant was repaired using 1.8-mm Mitek Mini-GII suture anchors. When the CFL remnant was of poor quality or absent, a Y-shaped tunnel configuration was created. A fibular tunnel was drilled at the common fibular insertion site of the ATFL and CFL (Figure 2), while separate talar and calcaneal tunnels were drilled at the respective anatomic insertion centers of the ATFL and CFL. The tendon graft was passed through the prepared tunnels with the assistance of a guide pin and secured in a neutral ankle position using PEEK (polyether ether ketone) interference screws (5-mm Milagro; Johnson & Johnson).

Postoperative Rehabilitation

After the surgery, the ankle was immobilized with short leg cast and the patient kept nonweightbearing for the first 2 weeks. Partial weightbearing was allowed at postoperative weeks 2 to 4 and gradually transitioned to full weightbearing with a walking brace. From week 6, weightbearing muscle strength exercise was allowed, and dynamic balance exercise started at week 7. All patients returned to full weightbearing walk and sport based on their tolerance. Then, patients were able to resume their daily activities and engage in sports that fell within their physical capabilities. Participating in high-demand sports was allowed at

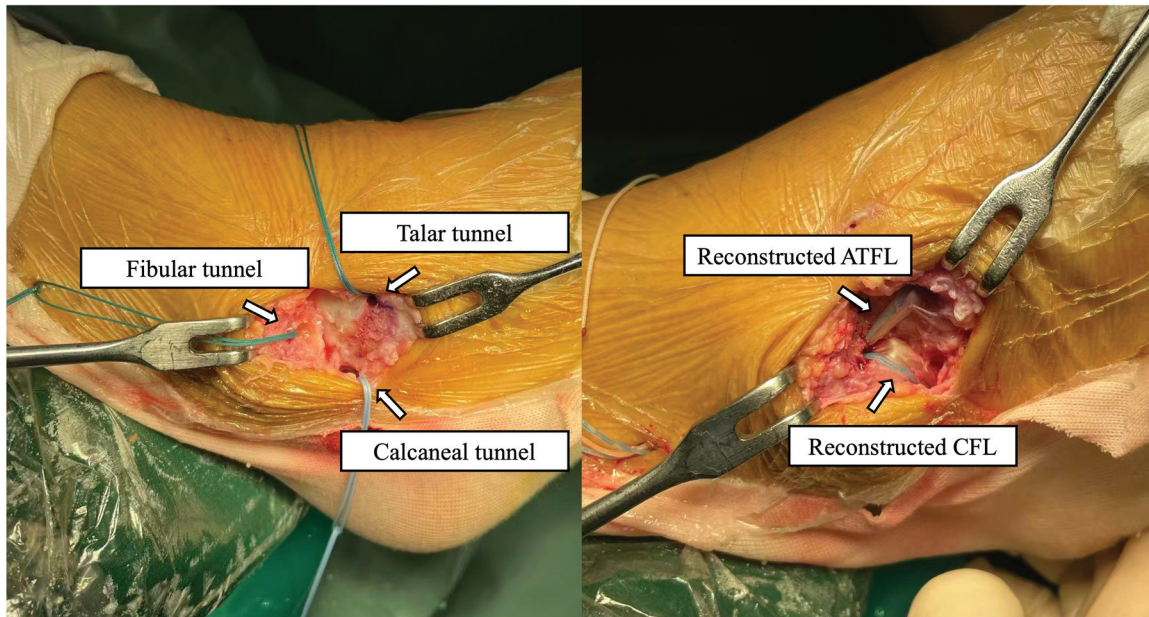


Figure 2. Intraoperative pictures. ATFL, anterior talofibular ligament; CFL, calcaneofibular ligament.

week 20. Routine follow-up was conducted at 2 weeks, 6 weeks, 3 months, and 6 months postoperatively.

Outcome Evaluation

Demographic characteristics including age at surgery, height, weight, body mass index (BMI), postinjury duration, preinjury Tegner Activity Level scale score, and smoking habits were collected from the patients' medical records. Intraoperative findings including concomitant CFL reconstruction, presence of os subfibulare, cartilage lesions, osteophytes, and syndesmosis width were obtained from operative documentation. Patient with a syndesmosis width ≥ 4 mm was determined as syndesmosis widening.³⁴

Clinical outcomes were assessed preoperatively and at the final follow-up using the visual analog scale, Tegner, and Karlsson-Peterson ankle score.³³ Resprain was defined as inversion injuries with lateral pain requiring ≥ 3 days for recovery. To minimize confounding, only noncontact resprain was recorded, and the patient was asked to recall the exact time of the first postoperative resprain as accurately as possible. Postoperative instability was identified when the patient had multiple resprains and a sense of instability. The occurrence of other postoperative complications, including range of motion restriction, infection, poor wound healing, incision numbness, donor-site morbidity, and knot pain were also recorded. In addition, whether the patient had returned to recreational sports and preinjury sports was also collected. Return to recreational or preinjury sports was defined as fully capable of participating at recreational level or the preinjury level.¹² Recreational sports were defined as cycling or other sports activities with a Tegner activity score ≥ 4 .³⁵ Patients with preinjury Tegner score ≥ 6 was determined as high sports demand.²¹

At our institution, patients routinely complete preoperative subjective scoring questionnaires 1 day before surgery, and the consent of willingness to be followed up postoperatively was signed. At the final follow-up, the researchers obtained patients' consent for study inclusion via telephone follow-up. The questionnaires and written informed consent were then sent to patients who were willing to engage.

Statistical Analysis

All the data were analyzed by IBM SPSS Statistics (Version 24.0; SPSS Inc). The Shapiro-Wilk test and Kolmogorov-Smirnov test was first performed to check for normal distribution. The data were presented as the mean $\pm$ SD for continuous variables, and frequency (percentage) for categorical data. The Pearson chi-square test, Fisher exact test, Mann-Whitney *U* test, or independent *t* test (2-sided test and $\alpha = .05$) was applied to compare the differences between groups. Differences were considered to be significant at $P < .05$.

A univariate logistic regression analysis was first performed to investigate potential factors influencing postoperative noncontact resprain. According to previous study by Park et al,²⁷ factors including sex, age at surgery > 30 years (the median age of the entire cohort), BMI > 25 kg/m², postinjury duration > 24 months, GJL, syndesmosis widening, high sports demand, CFL reconstruction, os subfibulare, osteochondral lesion of the talus, and presence of osteophytes were included in the univariate logistic regression. All factors with $P < .20$ in the univariate analysis were further included in the multivariate logistic regression to analyze the risk factors of postoperative noncontact resprain.⁴¹

TABLE 1
Patient Demographics of All Enrolled Patients^a

	All Patients (N = 290)	GJL (n = 72)	No GJL (n = 218)	P
Sex, male/female	156 (53.8) / 134 (46.2)	27 (37.5) / 45 (62.5)	129 (59.2) / 89 (40.8)	.001
Age at surgery, y	31.5 ± 10.2	27.5 ± 9.0	32.8 ± 10.2	<.001
BMI, kg/m ²	24.2 ± 3.3	23.8 ± 3.8	24.3 ± 3.1	.68
Postinjury duration, mo	51.1 ± 72.4	50.7 ± 73.0	51.2 ± 72.4	.25
Beighton score	2.7 ± 2.7	6.9 ± 1.4	1.4 ± 1.3	<.001
Syndesmosis width, mm	2.6 ± 1.3	2.8 ± 1.2	2.5 ± 1.3	.03
Preinjury Tegner score	5.8 ± 0.8	5.7 ± 0.8	5.8 ± 0.9	.94
CFL reconstruction, yes/no	134 (46.2) / 156 (53.8)	34 (47.2) / 38 (52.8)	100 (45.9) / 118 (54.1)	.84
Os subfibulare, yes/no	80 (27.6) / 210 (72.4)	24 (33.3) / 48 (66.7)	56 (25.7) / 162 (74.3)	.21
Osteochondral lesion of the talus, yes/no	100 (34.5) / 190 (65.5)	23 (31.9) / 49 (68.1)	77 (35.3) / 141 (64.7)	.60
Osteochondral lesion of the tibia, yes/no	30 (10.3) / 260 (89.7)	10 (13.9) / 62 (86.1)	20 (9.2) / 198 (90.8)	.26
Osteophyte, yes/no	108 (37.2) / 182 (62.8)	29 (40.3) / 43 (59.7)	79 (36.2) / 139 (63.8)	.54
Smoking habit, yes/no	13 (4.5) / 277 (95.5)	4 (5.6) / 68 (94.4)	9 (4.1) / 209 (95.9)	.74
Follow-up duration, mo	36.0 ± 7.1	36.8 ± 7.2	35.7 ± 7.1	.59

^aData are presented as mean ± SD or n (%). BMI, body mass index; CFL, calcaneofibular ligament; GJL generalized joint laxity. Bold values indicate significant difference between groups.

TABLE 2
Clinical Outcomes of All Enrolled Patients^a

	All Patients (N = 290)		P
	Preoperative	Postoperative	
Patient-reported outcome measure			
VAS for pain	4.0 ± 1.1	0.6 ± 0.9	<.001
Tegner score	3.6 ± 0.9	5.5 ± 1.0	<.001
Karlsson score	68.2 ± 8.3	94.2 ± 5.5	<.001
Postoperative resprain at 24 months follow-up, yes/no	12 (4.1) / 278 (95.9)		
Postoperative resprain, yes/no	19 (6.6) / 271 (93.4)		
Postoperative instability, yes/no	5 (1.7) / 285 (98.3)		
Return to recreational sports, yes/no	288 (99.3) / 2 (0.7)		
Return to preinjury sports, yes/no	216 (74.5) / 74 (25.5)		
Other surgical complications, yes/no	37 (12.8) / 253 (87.2)		

^aData are presented as mean ± SD or n (%). Bold values indicate significant difference between groups. VAS, visual analog scale.

PASS Sample Size software Version 15 (NCSS) was used for sample size calculation. Since no previous studies have investigated the risk factors of postoperative resprain following anatomic reconstruction with an autograft for CLAI patients, we calculated the achieved power of the study based on the difference between postoperative resprain rate of the patients with and without GJL. With an α set at .05, the study achieved a power of 0.841.

RESULTS

Patient Demographics and Clinical Outcomes of All Included Patients

Table 1 shows the patient demographics of all included patients, with a mean age of 31.5 ± 10.2 years and

a mean follow-up duration of 36.0 ± 7.1 months. Among them, 134 patients underwent CFL reconstruction, 136 patients underwent CFL repair, and 20 patients had an uninjured CFL. As shown in Table 2, at the final follow-up, noncontact resprain occurred in 19 of 290 patients (6.6%), and recurrent instability was observed in 5 of 290 patients (1.7%). At the 24-month follow-up, resprains occurred in 12 of 290 patients (4.1%). Compared with preoperative self-reported function, CLAI patients who underwent anatomic reconstruction with an autograft demonstrated significantly lower VAS scores (0.6 ± 0.9 vs 4.0 ± 1.1; $P < .001$) and significantly higher Tegner score (5.5 ± 1.0 vs 3.6 ± 0.9; $P < .001$), as well as Karlsson score (94.2 ± 5.5 vs 68.2 ± 8.3; $P < .001$) postoperatively. A total of 99.3% of patients returned to recreational sports, while 74.5% returned to their preinjury level of sports. Other surgical complications occurred in 12.8% of patients, and the

TABLE 3
Comparison of Clinical Outcomes of Patients With and Without GJL^a

	GJL (n = 72)	No GJL (n = 218)	P
VAS score			
Preoperative	4.1 ± 1.1	4.0 ± 1.1	.43
Postoperative	0.7 ± 1.0	0.5 ± 0.8	.27
Tegner score			
Preoperative	3.7 ± 0.9	3.6 ± 0.9	.96
Postoperative	5.4 ± 0.9	5.5 ± 1.0	.47
Karlsson score			
Preoperative	67.6 ± 10.1	68.3 ± 7.7	.58
Postoperative	90.0 ± 6.2	95.6 ± 4.4	<.001
Resprain, postoperative, 24 mo, yes/no	7 (9.7) / 65 (90.3)	5 (2.3) / 213 (97.7)	.01
Resprain, final follow-up	11 (15.3) / 61 (84.7)	8 (3.7) / 210 (96.3)	.001
Recurrent instability, postoperative, yes/no	5 (6.9) / 67 (93.1)	0 (0.0) / 218 (100.0)	.001
Return to recreational sports, yes/no	72 (100.0) / 0 (0.0)	216 (99.1) / 2 (0.9)	≥.99
Return to preinjury sports, yes/no	51 (70.8) / 21 (29.2)	165 (75.7) / 53 (24.3)	.41
Other surgical complications, yes/no	9 (12.5) / 63 (87.5)	28 (12.8) / 190 (87.2)	.94

^aData are presented as mean ± SD or n (%). Bold values indicate significant difference between groups. GJL, generalized joint laxity; VAS, visual analog scale.

most common surgical complications were ROM restriction (6.2%) and donor site morbidity (3.8%).

Demographic Comparison of Patients With and Without GJL

As shown in Table 1, the significant demographics and intra-articular lesion associated with generalized ligamentous laxity occurred in a low proportion of male patients (37.5% vs 62.5%; $P = .001$), a younger age at surgery (27.5 ± 9.0 vs 32.8 ± 10.2 ; $P < .001$), and a larger syndesmosis width (2.8 ± 1.2 vs 2.5 ± 1.3 ; $P = .03$). There was no significant difference in the follow-up duration between the 2 groups.

Comparison of Clinical Outcomes of Patients With and Without GJL

As shown in Table 3, there were no significant differences in preoperative VAS score, Tegner score, and Karlsson score between the 2 groups. There was also no significant difference in the postoperative VAS score, Tegner score, return to sports recreational sports status, return to preinjury sports status, and the presence of other surgical complications between the 2 groups. The patients with GJL demonstrated a lower Karlsson score (90.0 ± 6.2 vs 95.6 ± 4.4 ; $P < .001$), a higher resprain rate at 24 months postoperatively (9.7% vs 2.3%; $P = .01$), a higher noncontact resprain rate at the final follow-up (15.3% vs 3.7%; $P = .001$), and a higher rate of postoperative instability (6.9% vs 0.0%; $P = .001$) compared to those without GJL. The mean time to first postoperative resprain after the reconstruction was 18.3 ± 8.9 months for patients in the GJL group and 23.9 ± 10.6 months for patients in the non-GJL group.

Analysis of Potential Factors Influencing Postoperative Resprain

Due to the low overall prevalence of smoking and the low presence of osteochondral lesions of the tibia in the cohort, the 2 factors are excluded from the univariate logistic regression analyses. As shown in Table 4, based on the result of univariate logistic regression, 3 potential factors with $P < .20$ including GJL, syndesmosis widening, and participating in high-demand sport were identified. As shown in Table 5, the result of multivariate logistic regression demonstrated that GJL ($P = .004$; odds ratio = 4.293; 95% CI, 1.612-11.431) is an independent predictor of postoperative resprain. Besides, participating in high-demand sports was also an independent predictor of postoperative resprain ($P = .04$; odds ratio = 3.325; 95% CI, 1.043-10.600).

Comparison of Cumulative Rate of Postoperative Resprain in Patients With and Without GJL

As presented in Figure 3, Kaplan-Meier analysis with log-rank testing demonstrated that the estimations of the non-resprain rate at the final follow-up in CLAI patients with GJL was significantly lower than patients without GJL ($P = .001$).

DISCUSSION

The main finding of our study was that GJL can lead to a significantly higher postoperative noncontact resprain rate in CLAI patients following anatomic reconstruction using an autograft. The result of the multivariate logistic regression suggests that GJL and high-demand sports are independent predictors of postoperative noncontact

TABLE 4
Results of Univariate Logistic Regression Analysis for Identifying Factors
Influencing Postoperative Resprain^a

	<i>P</i>	Odds Ratio (95% CI)
Male sex	.62	1.344 (0.416-4.342)
Age at surgery >30 y	.20	0.467 (0.145-1.507)
BMI >25 kg/m ²	.90	1.079 (0.325-3.584)
Postinjury duration >24 mo	.65	0.782 (0.274-2.234)
GJL	.006	4.350 (1.526-12.402)
Syndesmosis widening	.12	2.382 (0.809-7.010)
High sports demand	.06	3.175 (0.941-10.714)
CFL reconstruction	.67	1.257 (0.440-3.589)
Os subfibulare	.27	1.903 (0.606-5.973)
Osteochondral lesion of the talus	.53	0.661 (0.181-2.413)
Osteophyte	.36	0.562 (0.166-1.907)

^aBold value indicates significant difference between groups. BMI body mass index; CFL, calcaneofibular ligament; GJL, generalized joint laxity.

TABLE 5
Results of Multivariate Logistic Regression Analysis for Identifying Factors
Influencing Postoperative Resprain^a

	<i>P</i>	Adjusted Odds Ratio (95% CI)
GJL	.004	4.293 (1.612-11.431)
Syndesmosis widening	.05	2.675 (0.993-7.206)
High sports demand	.04	3.325 (1.043-10.600)

^aBold values indicate significant difference between groups. GJL, generalized joint laxity.

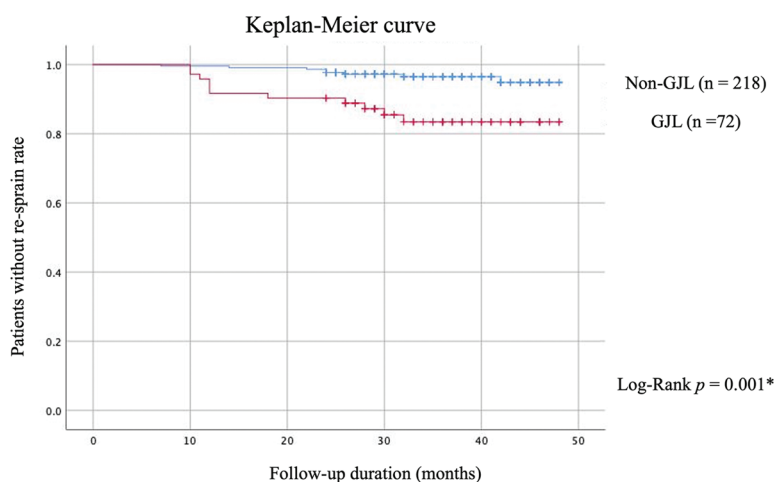


Figure 3. Kaplan-Meier analysis with log-rank test showing the cumulative rate of postoperative resprain in patients with and without generalized joint laxity (GJL). Asterisk indicates significant difference between groups.

resprain of CLAI patients following anatomic reconstruction using an autograft.

The overall noncontact resprain rate of anatomic reconstruction with an autograft was 6.6%, while the noncontact resprain rate in patients without GJL was 3.7%.

Multivariate analysis revealed that there was no significant association between established risk factors for modified Broström repair failure such as obesity and postoperative resprain, indicating excellent outcomes. Clanton et al⁶ have found that anatomic reconstruction

demonstrated similar strength and stiffness to the intact ATFL at time zero in a cadaveric model. Furthermore, the study by Prisk et al³⁰ demonstrated that anatomic reconstruction better restores ankle joint contact mechanics and hindfoot motion patterns compared with the modified Broström repair. These biomechanical findings, along with extensive clinical evidence, indicate that despite requiring longer postoperative recovery before return to sports, anatomic reconstruction can provide excellent superior ankle stability.^{9,14,35}

In our study, patients with GJL following anatomic reconstruction demonstrated a noncontact resprain rate of 15.3% at a mean follow-up of 36.8 months, while the resprain rate at the 24-month follow-up was 9.7%. Multivariate logistic regression identified that GJL is an independent predictor of postoperative resprains. Previous studies suggest that this may be attributed to impaired collagen synthesis in patients with GJL,^{17,19} which can lead to increased laxity of the ATFL under internal rotational force.^{46,47} Numerous previous studies have regarded GJL as a contraindication for the modified Broström procedure.^{27,45} In studies focusing on shorter-term clinical effectiveness, Wang et al⁴⁰ reported a 19.7% failure rate at 24 months postoperatively among 61 GJL patients undergoing a modified Broström procedure, while Huang et al¹³ reported a 28.1% resprain rate in 32 GJL patients at a mean follow-up of 27.4 months.

Meanwhile, recent studies suggest there remains potential for optimizing treatment strategies for CLAI patients with GJL. Analyzing the signal-to-noise value of the reconstructed ATFL on MRI imaging, Cordier et al⁷ reported that the ligamentization process of the graft can last for 1 to 2 years postoperatively, which was supported by other recent studies.^{2,29} Considering that CLAI patients who undergo anatomic reconstruction typically initiate sports participation at 16 weeks and gradually return to preinjury activity levels between 6 and 9 months postoperatively,^{22,28} this indicates that biomechanical strength recovery of the reconstructed ligament may extend beyond the typical return-to-sport timeline in clinical practice. In our study, over a quarter of the noncontact resprains in CLAI patients with GJL occurred during participation in sports activities between 6 months and 1 year postoperatively. This suggests that the prolonged rehabilitation strategies, cautious return-to-sport decisions within 1 year postoperatively, and bracing for a period of time after return to sports for CLAI patients with GJL may require additional caution to allow sufficient biomechanical maturation of the graft, thereby potentially reducing the resprain rate, especially for patients participating in high-demand sports.¹¹ Regarding the prevention of resprain >12 months postoperatively, studies focusing on MRI findings after ACL reconstruction suggest that another potentially effective method to reduce postoperative resprain is to incorporate routine MRI examination into postoperative follow-up.⁸ The study by Putnis et al³¹ demonstrated that MRI at 12 months postoperatively can identify those at higher risk of graft failure following hamstring tendon autograft ACL reconstruction, with a sensitivity of 66% and a specificity of 77%, while a recent

study by Lutz et al²⁵ similarly found that patients who experienced graft retear >12 months postoperatively exhibited higher ligament signal-to-noise values at 12 months postoperatively. Therefore, further research is needed to see if MRI can ascertain healing of the ATFL graft and help determine return to sports. Suture tape augmentation also appears to be a potential alternative optimization approach. In biomechanical studies, suture tape augmentation can provide comparable initial strength compared with intact ATFL,^{3,20,43} and the synthetic nature of the augmented suture tape theoretically mitigates the potential of postoperative laxity in GJL patients; indeed it has been adopted by some institutions as an optional surgical procedure for CLAI patients with poor ligament remnant, GJL, or participation in high-demand sports.^{4,38} For cohorts and case series focusing on the clinical efficiency of GJL patients following suture tape augmentation, Cho et al⁵ reported that only 3.6% of the CLAI patients with GJL following suture tape augmentation demonstrated recurrent instability at the 24-month follow-up, while a recent study by Qu et al³² reported that there were no significant differences in postoperative talar tilt angle or anterior translation of the talus between CLAI patients with and without GJL following suture tape augmentation at mean follow-up of 21.7 months. However, it should be noted that current evidence regarding suture tape augmentation for CLAI with GJL still remains incomplete. Most of the previous studies were retrospective with small sample sizes and have not directly compared the impact of suture tape augmentation on the postoperative resprain in CLAI patients with and without GJL. Therefore, larger-scale, higher level evidence studies are warranted to further investigate the clinical efficacy of suture tape augmentation for CLAI patients with GJL.

To our knowledge, this is the first study to investigate the effect of GJL on the clinical outcomes in CLAI patients following anatomic reconstruction using an autograft. Our findings suggest that GJL is an independent predictor of resprain following anatomic reconstruction with an autograft, which might increase postoperative laxity. Therefore, prolonged postoperative rehabilitation strategies, cautious return-to-sport decisions, and incorporation of routine MRI examination can be potential approaches to reduce postoperative resprain of CLAI patients with GJL. Anatomic reconstruction using an allograft or using suture tape augmentation might be a potential alternative but requires further study to investigate the clinical efficiency of these approaches.

Limitations

There are some limitations to this study. First, as a retrospective study, selection bias can exist, and the decision on CFL reconstruction or CFL repair or untreated CFL was made at the surgeons' discretion, which represents another bias. Furthermore, the presence of resprain was retrospectively collected at the final follow-up. Therefore, our findings may be influenced by recall bias regarding sprain recurrence and timing. Additionally, though our study

defined resprain as inversion injuries requiring ≥ 3 days to recover, the severity of the resprain was ungraded and restricted us from further analysis. Furthermore, we did not investigate the association between anatomical (eg, tibiotalar sector) or surgical (eg, bone tunnel position, graft bend angle) factors and postoperative resprain, which were reported as risk factors of poor outcomes following modified Broström procedure or ACL reconstruction. Thus, prospective studies incorporating more detailed potential factors are needed to identify more comprehensive predictors of resprains for CLAI patients following anatomic reconstruction using an autograft.

CONCLUSION

GJL is an independent predictor of postoperative noncontact resprain of CLAI patients following anatomic reconstruction using an autograft.

ORCID iDs

Qinwei Guo  <https://orcid.org/0000-0003-3600-9544>

Dong Jiang  <https://orcid.org/0000-0003-4380-7683>

REFERENCES

- Balduini FC, Vegso JJ, Torg JS, Torg E. Management and rehabilitation of ligamentous injuries to the ankle. *Sports Med*. 1987;4(5):364-380.
- Bilichtin E, Rougureau G, Rollet ME, et al. MRI evaluation of ATFL and CFL ligamentization after anatomical surgical reconstruction with a hamstring graft. *Foot Ankle Surg*. 2025;31(1):74-78.
- Boey H, Verfaillie S, Natsakis T, Vander Sloten J, Jonkers I. Augmented ligament reconstruction partially restores hindfoot and mid-foot kinematics after lateral ligament ruptures. *Am J Sports Med*. 2019;47(8):1921-1930.
- Cho BK, Kim SH. The modified Broström procedure with suture-tape augmentation for chronic lateral ankle instability. *J Clin Med*. 2025;14(5):1683.
- Cho B-K, Park K-J, Park J-K, SooHoo NF. Outcomes of the modified Broström procedure augmented with suture-tape for ankle instability in patients with generalized ligamentous laxity. *Foot Ankle Int*. 2017;38(4):405-411.
- Clanton TO, Viens NA, Campbell KJ, Laprade RL, Wijdicks CA. Anterior talofibular ligament ruptures, part 2: biomechanical comparison of anterior talofibular ligament reconstruction using semitendinosus allografts with the intact ligament. *Am J Sports Med*. 2014;42(2):412-416.
- Cordier G, Boudahmane S, Ovigie J, Michels F, Nunes GA, Dallaudiere B. MRI assessment of tendon graft after lateral ankle ligament reconstruction: does ligamentization exist? *Am J Sports Med*. 2024;52(3):721-729.
- D'Ambrosi R, Sconfienza LM, Albano D, Meena A, Abermann E, Fink C. Can MRI predict return to sport after anterior cruciate ligament reconstruction? A systematic review of the literature. *Radiol Med*. 2025;130(5):638-649.
- Dierckman BD, Ferkel RD. Anatomic reconstruction with a semitendinosus allograft for chronic lateral ankle instability. *Am J Sports Med*. 2015;43(8):1941-1950.
- Gribble PA, Delahunt E, Bleakley C, et al. Selection criteria for patients with chronic ankle instability in controlled research: a position statement of the International Ankle Consortium. *J Orthop Sports Phys Ther*. 2013;43(8):585-591.
- Hardy A, Freiha K, Moussa MK, et al. Use of Ankle-GO to assess and predict return to sport after lateral ankle reconstruction for chronic ankle instability. *Orthop J Sports Med*. 2025;13(3):23259671251322903.
- Hou Z-C, Su T, Ao Y-F, et al. Arthroscopic modified Broström procedure achieves faster return to sports than open procedure for chronic ankle instability. *Knee Surg Sports Traumatol Arthrosc*. 2022;30(10):3570-3578.
- Huang B, Kim YT, Kim JU, Shin JH, Park YW, Kim HN. Modified Broström procedure for chronic ankle instability with generalized joint hypermobility. *Am J Sports Med*. 2016;44(4):1011-1016.
- Kennedy JG, Smyth NA, Fansa AM, Murawski CD. Anatomic lateral ligament reconstruction in the ankle: a hybrid technique in the athletic population. *Am J Sports Med*. 2012;40(10):2309-2317.
- Kim SJ, Chang JH, Oh KS. Posterior cruciate ligament reconstruction in patients with generalized joint laxity. *Clin Orthop Relat Res*. 2009;467(1):260-266.
- Kim S-J, Choi CH, Lee S-K, Lee W, An H, Jung M. Minimum two-year follow-up of anterior cruciate ligament reconstruction in patients with generalized joint laxity. *J Bone Joint Surg Am*. 2018;100(4):278-287.
- Kim SJ, Kim TE, Lee DH, Oh KS. Anterior cruciate ligament reconstruction in patients who have excessive joint laxity. *J Bone Joint Surg Am*. 2008;90(4):735-741.
- Kim S-J, Moon H-K, Kim S-G, Chun Y-M, Oh Y-S. Does severity or specific joint laxity influence clinical outcomes of anterior cruciate ligament reconstruction? *Clin Orthop Relat Res*. 2010;468(4):1136-1141.
- Krebs NM, Barber-Westin S, Noyes FR. Generalized joint laxity is associated with increased failure rates of primary anterior cruciate ligament reconstructions: a systematic review. *Arthroscopy*. 2021;37(7):2337-2347.
- Larkins CG, Brady AW, Aman ZS, Dorman GJ, Haytmanek CT, Clanton TO. Evaluation of the intact anterior talofibular and calcaneofibular ligaments, injuries, and repairs with and without augmentation: a biomechanical robotic study. *Am J Sports Med*. 2021;49(9):2432-2438.
- Li X, Killie H, Guerrero P, Busconi BD. Anatomical reconstruction for chronic lateral ankle instability in the high-demand athlete: functional outcomes after the modified Broström repair using suture anchors. *Am J Sports Med*. 2009;37(3):488-494.
- Li Y, Su T, Hu Y, et al. Return to sport after anatomic lateral ankle stabilization surgery for chronic ankle instability: a systematic review and meta-analysis. *Am J Sports Med*. 2024;52(2):555-566.
- Lopes R, Hong CC, Calder J, Kerkhoffs GMMJ. Risk factors for the recurrence of instability after operative treatment of chronic lateral ankle instability: a systematic review. *J Exp Orthop*. 2025;12(1):e70214.
- Luthfi A, Nakasa T, Ikuta Y, et al. Risk factors for the recurrence of instability after lateral ankle ligament repair. *Foot Ankle Int*. 2023;44(7):617-628.
- Lutz C, Mancino F, Parker DA. Twelve-month magnetic resonance imaging after anterior cruciate ligament reconstruction can identify risk factors for subsequent graft rupture and used to guide the return to sport. Results from a high-volume institution. *J ISAKOS*. 2025;14:100994.
- Malfait F, Francomano C, Byers P, et al. The 2017 international classification of the Ehlers-Danlos syndromes. *Am J Med Genet C Semin Med Genet*. 2017;175(1):8-26.
- Park KH, Lee JW, Suh JW, et al. Generalized ligamentous laxity is an independent predictor of poor outcomes after the modified Broström procedure for chronic lateral ankle instability. *Am J Sports Med*. 2016;44(11):2975-2983.
- Pearce CJ, Tourné Y, Zellers J, Terrier R, Toschi P, Silbernagel KG; ESKKA-AFAS Ankle Instability Group. Rehabilitation after anatomical ankle ligament repair or reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 2016;24(4):1130-1139.

29. Philippe C, Zrounba V, Lopes R, Cavaignac E, Ancelin D. MRI evaluation of graft maturation following arthroscopic anatomical lateral ligament reconstruction: a one-year follow-up study. *J Foot Ankle Surg.* 2025;64(6):722-728.
30. Prisk VR, Imhauser CW, O'Loughlin PF, Kennedy JG. Lateral ligament repair and reconstruction restore neither contact mechanics of the ankle joint nor motion patterns of the hindfoot. *J Bone Joint Surg Am.* 2010;92(14):2375-2386.
31. Putnis SE, Oshima T, Klasan A, et al. Magnetic resonance imaging 1 year after hamstring autograft anterior cruciate ligament reconstruction can identify those at higher risk of graft failure: an analysis of 250 cases. *Am J Sports Med.* 2021;49(5):1270-1278.
32. Qu F, Ji L, Sun C, et al. Arthroscopic anterior talofibular ligament repair combined with all-inside suture tape augmentation for treatment of chronic lateral ankle instability with generalized joint laxity. *Foot Ankle Int.* 2024;45(10):1102-1110.
33. Rupp MC, Degenhardt H, Winkler PW, et al. High return to sports and return to work rates after anatomic lateral ankle ligament reconstruction with tendon autograft for isolated chronic lateral ankle instability. *Knee Surg Sports Traumatol Arthrosc.* 2022;30(11):3862-3870.
34. Su T, Du MZ, Parekh SG, et al. Effect of arthroscopically confirmed syndesmotomic widening on outcome following isolated Broström operation for chronic lateral ankle instability. *Foot Ankle Int.* 2023;44(4):270-278.
35. Su T, Wang A-H, Guo Q-W, et al. Both open and arthroscopic all-inside anatomic reconstruction with autologous gracilis tendon restore ankle stability in patients with chronic lateral ankle instability. *Arthroscopy.* 2023;39(4):1035-1045.
36. Takao M, Oae K, Uchio Y, Ochi M, Yamamoto H. Anatomical reconstruction of the lateral ligaments of the ankle with a gracilis autograft: a new technique using an interference fit anchoring system. *Am J Sports Med.* 2005;33(6):814-823.
37. Vega J, Malagelada F, Dalmau-Pastor M. Arthroscopic all-inside ATFL and CFL repair is feasible and provides excellent results in patients with chronic ankle instability. *Knee Surg Sports Traumatol Arthrosc.* 2020;28(1):116-123.
38. Vermorel PH, Vega J, Guelfi M. Suture tape augmentation in arthroscopic lateral ligament repair for chronic ankle instability yields similar clinical outcomes but faster return to sport compared to isolated repair. *Knee Surg Sports Traumatol Arthrosc.* 2026;34(4):1442-1451.
39. Vuurberg G, Pereira H, Blankevoort L, van Dijk CN. Anatomic stabilization techniques provide superior results in terms of functional outcome in patients suffering from chronic ankle instability compared to non-anatomic techniques. *Knee Surg Sports Traumatol Arthrosc.* 2018;26(7):2183-2195.
40. Wang AH, Su T, Jiang YF, et al. Arthroscopic modified Broström procedure achieved similar favorable short term outcomes to open procedure for chronic lateral ankle instability cases with generalized joint laxity. *Knee Surg Sports Traumatol Arthrosc.* 2023;31(9):4043-4051.
41. Wang DY, Jiao C, Ao YF, et al. Risk factors for osteochondral lesions and osteophytes in chronic lateral ankle instability: a case series of 1169 patients. *Orthop J Sports Med.* 2020;8(5):2325967120922821.
42. Waterman BR, Owens BD, Davey S, Zacchilli MA, Belmont PJ Jr. The epidemiology of ankle sprains in the United States. *J Bone Joint Surg Am.* 2010;92(13):2279-2284.
43. Willegger M, Benca E, Hirtler L, et al. Biomechanical stability of tape augmentation for anterior talofibular ligament (ATFL) repair compared to the native ATFL. *Knee Surg Sports Traumatol Arthrosc.* 2016;24(4):1015-1021.
44. Won SH, Yi Y, Cho J, et al. Ligament quality predicts recurrence and functional outcomes after arthroscopic all-inside anterior talofibular ligament repair. *Arthroscopy.* 2025;41(11):4740-4749.
45. Xu HX, Lee KB. Modified Broström procedure for chronic lateral ankle instability in patients with generalized joint laxity. *Am J Sports Med.* 2016;44(12):3152-3157.
46. Yokoe T, Tajima T, Kawagoe S, Yamaguchi N, Morita Y, Chosa E. The ratio of stress to nonstress anterior talofibular ligament length on ultrasonography: normative values. *Orthop J Sports Med.* 2021;9(11):23259671211056305.
47. Yokoe T, Tajima T, Yamaguchi N, Morita Y, Chosa E. The anterior talofibular ligament ratio was greater in young men with generalized joint laxity than in those without generalized joint laxity. *Knee Surg Sports Traumatol Arthrosc.* 2023;31(5):1994-2000.